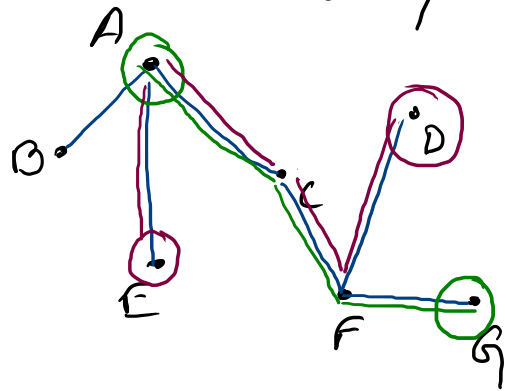
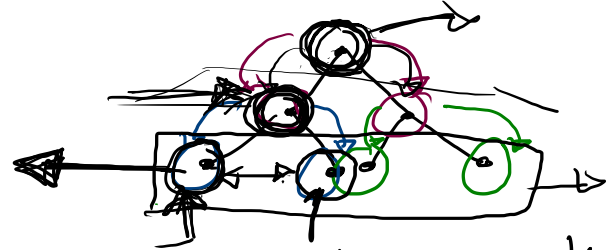


Chapter 11.1 Trees.

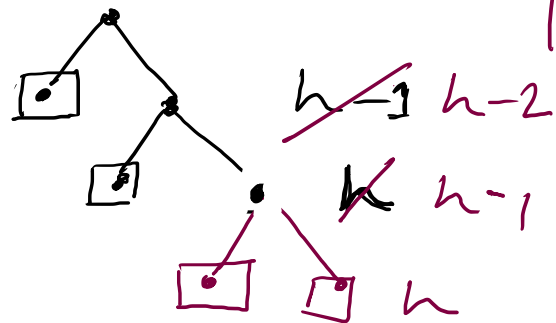
Trees → CG with no simple circuit. $m = n - 1$

→ will be a tree iff there is a unique path b/w any 2 of its vertices.



* Rooted tree

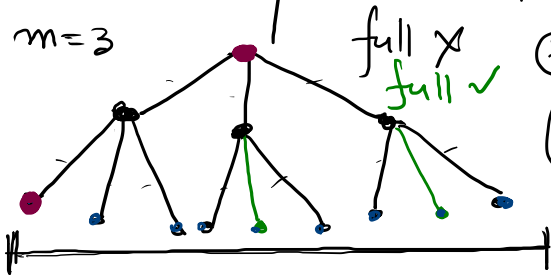
→ one node assigned as the root/starting point and the edges are directed away from root.



m -ary trees. → if every internal vertices has no more than m children

full m -ary trees → every internal vertices has exactly m children.

Properties of trees.



$$n=13 \quad m=3$$

$$l=12=n-1$$

$$l=9 \quad iV=3$$

$$h=2$$

$$l=9=3^2=m^h$$

$$h=\lceil \log_m l \rceil$$

$$= \lceil \log_3 9 \rceil$$

$$h=2$$

① $\Rightarrow$ A tree with n -vertices has $(n-1)$ edges.

② A full m -ary tree

a/ n -vertices $\rightarrow i = (n-1)/m \rightarrow$ internal vertices
 $\rightarrow l = \lceil (m-1)i + 1 \rceil / m \rightarrow$ leaves.

b/ $iV \rightarrow n = mi + 1 \rightarrow$ nodes.
 $\rightarrow l = (m-1)i + 1 \rightarrow$ leaves.

c/ l leaves $\rightarrow n = (ml-1)/(m-1) \rightarrow$ vertices.
 $\rightarrow i = (l-1)/(m-1) \rightarrow iV$

⑩ level of a node

↳ length of path from 'v' to root
↳ level of root $\rightarrow 0$

⑪ height of a tree $\rightarrow \forall n \max(\text{level}_n)$

⑫ m-ary tree of height h

↳ balanced/unbalanced?

↳ if all the leaves of the tree are in levels h or (h-1)

↳ m-ary tree of height h $\rightarrow$ can have m^h leaves.

↳ m-ary tree $\begin{cases} \text{height } h \\ \text{leaves } l \end{cases} \rightarrow \begin{cases} \text{not full} & h \geq \lceil \log_m l \rceil \\ \text{balanced} & h = \lceil \log_m l \rceil \\ \text{full} & \end{cases}$